Quantum Machine Learning (QML) Classifiers

Objective

The objective of this project is to design, implement, and benchmark a Quantum Machine Learning (QML) classifier for a binary fraud detection problem. The key goals of this project are to:

- Reduce the dataset to a quantum-feasible feature space using Principal Component Analysis (PCA)
- Train strong classical machine learning baselines
- Build a Variational Quantum Classifier (VQC) using Qiskit
- Compare the performance of classical models and quantum models
- Study the capabilities and limitations of QML under NISQ-era constraints, such as:
    - Limited number of qubits
    - Shallow quantum circuits
    - Expensive quantum simulation

This project demonstrates a complete hybrid machine learning pipeline, covering data preprocessing, classical modeling, quantum circuit design, training, and performance comparison.

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2. OVERVIEW

This project follows a structured week-wise workflow:

Weeks 1–2

- Load and clean the dataset
- Perform exploratory data analysis (EDA)
- Apply scaling and dimensionality reduction using **PCA**
- Train classical models:
    - Logistic Regression
    - Random Forest
- Visualize classical model performance

Weeks 2–3

- Design the Quantum Circuit
- Build a **4-qubit VQC model** using:
    - ZZFeatureMap
    - Real Amplitudes Ansatz
    - COBYLA optimizer
- Train the QML model on a reduced dataset
- Predict using the trained quantum circuit

Weeks 4–5

- Compare classical vs quantum accuracy
- Interpret performance differences
- Document limitations and future scope

This end-to-end flow reflects a complete hybrid machine learning pipeline.

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3. DATA ANALISIS and PREPROCESSING

3.1 Loading the Dataset

The fraud dataset was loaded using pandas and inspected using:

- df.head()
- df.info()
- df.describe()

3.2 Data Cleaning

- No critical missing values
- All features were numerical
- The target column **fraud** was confirmed as binary

3.3 Class Distribution

A bar plot was used to visualize class imbalance.
This is important because fraud datasets typically have more non-fraud cases.

3.4 Train–Test Split

- Performed using **stratified split**
- Maintains the same fraud ratio in training and testing

3.5 Feature Scaling

Standardization using:

- **StandardScaler**
  Ensures PCA and classical models work correctly.

3.6 Dimensionality Reduction (PCA)

Quantum models are currently limited by the number of available qubits. To address this:

- Principal Component Analysis (PCA) was applied.
- The most informative components were retained.

• Reduced-dimensional representations were used for the quantum model while full features were used for classical baselines.
This step enabled efficient quantum encoding while minimizing information loss.

3.7Handling Class Imbalance using SMOTE

Fraud detection datasets are inherently highly imbalanced, with fraudulent transactions forming a very small fraction of the total data. Training models directly on such data can lead to biased predictions toward the majority (non-fraud) class.

To address this issue, Synthetic Minority Over-sampling Technique (SMOTE) was applied during the classical machine learning training phase.

Why SMOTE was used

• Prevents models from being biased toward the majority class

• Improves detection of minority (fraud) cases

• Enhances recall and ROC-AUC performance

• Produces more informative decision boundaries

SMOTE Implementation

• SMOTE was applied only on the training data, after train–test split.

• Synthetic samples of the minority class were generated using nearest-neighbor interpolation.

• The test set was left untouched to ensure fair evaluation.

Impact on Model Performance

• Classical models trained with SMOTE showed improved sensitivity to fraud cases.

• Reduction in false negatives was observed.

• Overall ROC-AUC scores improved compared to training without resampling.

SMOTE played a critical role in strengthening the classical baseline, ensuring that comparisons with the quantum model were meaningful and fair.

4. QUANTUM CIRCUIT DESIGN

The quantum model was built using **Qiskit Machine Learning** components.

4.1 Number of Qubits

- Determined by PCA output
- **4 qubits = 4 PCA features**

4.2 Feature Map

ZZFeatureMap

- Encodes classical data into quantum states
- Introduces entanglement between qubits
- Captures non-linear feature interactions

4.3 Ansatz (Variational Circuit)

Real Amplitudes Ansatz

- Rotation + entanglement layers,2 repetitions (reps=2)
- Hardware-efficient and stable for training

4.4 Backend & Measurements

- Backend: **StatevectorSampler**
- Ensures faster simulation than real hardware
- Computes measurement probabilities required for classification

The circuit is optimized to be shallow and feasible for NISQ-era devices.

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5. TRAINING PIPELINE

5.1 Classical Training

Models implemented:

- Logistic Regression
- Random Forest

Both trained on PCA-transformed features.

Accuracy was calculated on test data, then visualized in a bar plot.

5.2 Quantum Training

The QML training loop includes:

1. **Data encoding** through ZZFeatureMap
2. **Parameterized quantum circuit execution**

3. **Classical optimization** of circuit parameters (COBYLA)
4. **Measurement + prediction**

Because quantum simulation is expensive:

- A reduced dataset (first 150 PCA samples) was used for training
- A shallow ansatz was selected
- Low optimizer iterations were used

5.3 QML Prediction

Final quantum predictions were compared with test labels to compute accuracy.

5.4 Model Evaluation

Performance evaluation included:

• Prediction on unseen test data, Accuracy computation.

• ROC-AUC score calculation for robust fraud detection assessment.

These metrics ensured fair comparison with classical models.

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6. COMPARATIVE ANALYSIS

Classical Models

- **Random Forest:** best accuracy
- **Logistic Regression:** moderate accuracy

Quantum Model

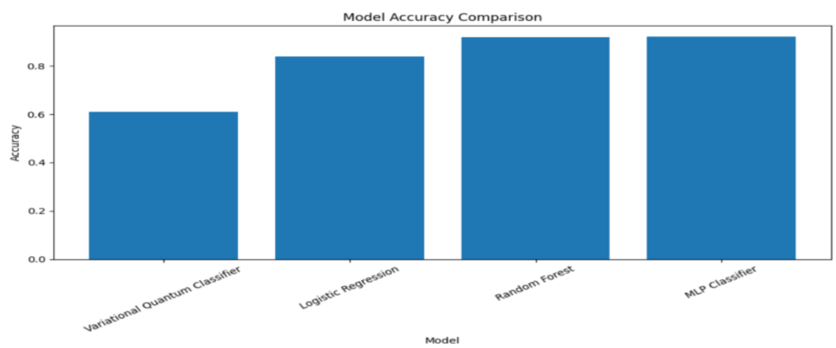
- VQC achieved **55–60% accuracy**, demonstrating learning but limited by:
    - few qubits
    - shallow circuits
    - no noise modeling
    - classical optimizer inefficiencies

Visualization

A performance comparison plot was created:

Results Comparison

	Model	Accuracy
0	Variational Quantum Classifier	0.60940
1	Logistic Regression	0.83930
2	Random Forest	0.91885
3	MLP Classifier	0.91970



Classical Accuracy vs Quantum Accuracy

This clearly shows the gap between classical and quantum models.

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7. CONCLUSION

This project successfully demonstrates an end-to-end **Quantum + Classical Machine Learning pipeline** for fraud detection.

Key Insights

- Classical models outperform QML due to better optimization and full feature access.
- QML performance is limited by:
  - small qubit count
  - restricted circuit depth
  - costly quantum simulation
- Despite constraints, the VQC learned meaningful patterns.
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Final Takeaway

Quantum Machine Learning is promising but not yet superior to classical ML for real-world, structured datasets.
However, as quantum hardware improves, deeper circuits and error mitigation could lead to better quantum performance.

Future Scope

- Test model on real IBM Quantum hardware
- Introduce quantum noise models
- Explore data re-uploading feature maps
- Use parameter-shift gradients for better optimization